

Chapter 9. Linear Approximation, Tangent Planes, and Differentials

The local linear model behind tangent planes, sensitivity, and uncertainty

The central message of this chapter is simple and powerful:

a differentiable multivariable function is approximately linear near a point.

In single-variable calculus, the tangent line is the best first-order approximation to a curve near a point. In multivariable calculus, the same idea becomes the *tangent plane* for a graph $z = f(x, y)$ and the *tangent hyperplane* for a function of many variables.

Algebraically, the approximation is built from partial derivatives. Geometrically, it is the flat object that best matches the surface near the point. Computationally, it is the first local model used in numerical methods, optimization, uncertainty propagation, and statistical approximation.

Core principle

A differentiable multivariable function is approximately linear near a point.

Teaching-material connection

This chapter develops tangent planes, linearization, differential approximation, differentiability, total differentials, and tangent planes to level surfaces. The examples $z = 3x^2 + y^2$, gradient-based approximation, and the level-surface problem $F(x, y, z) = x^3 + y^2e^z + xyz$ are used as core models.

Learning goals

After studying this chapter, you should be able to:

1. explain why tangent planes are multivariable tangent lines;
2. find the tangent plane to a surface $z = f(x, y)$ at a point;
3. compute the linearization $L(x, y)$ of a function $f(x, y)$ at (a, b) ;
4. use linearization to estimate nearby function values;
5. write linear approximation in gradient-dot-product form;
6. explain the difference between partial derivatives existing and differentiability;
7. compute total differentials and use them for error propagation;
8. find tangent planes to level surfaces $F(x, y, z) = c$;
9. interpret local linear models in statistics, data science, and machine learning;
10. use computation to visualize tangent planes and approximation errors.

1 Review: tangent lines and local linearity

For a single-variable function $y = f(x)$, the tangent line at $x = a$ is

$$L(x) = f(a) + f'(a)(x - a).$$

The tangent line is useful because when x is close to a ,

$$f(x) \approx L(x).$$

If $\Delta x = x - a$, then

$$\Delta y = f(x) - f(a) \approx f'(a)\Delta x.$$

The multivariable version is

$$\Delta z \approx f_x(a, b)\Delta x + f_y(a, b)\Delta y.$$

The right side is linear in the input changes Δx and Δy .

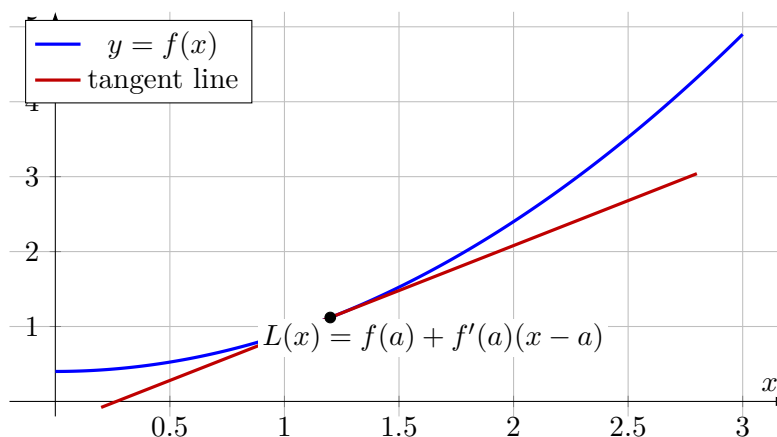


Figure 1: In one variable, a tangent line is the local linear model. In two variables, the tangent line becomes a tangent plane.

Local does not mean global

A tangent line or tangent plane usually gives a good approximation only near the point of tangency. Far away from the point, nonlinear curvature becomes important.

2 Tangent planes to graphs $z = f(x, y)$

Let $z = f(x, y)$ be differentiable near (a, b) , and let

$$P = (a, b, f(a, b))$$

be the point on the graph. The tangent plane at P is

$$z - f(a, b) = f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Equivalently,

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

The function on the right is the *linearization* of f at (a, b) :

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

The approximation

$$f(x, y) \approx L(x, y)$$

is called the *linear approximation* or *tangent plane approximation*.

At the point (a, b) , the tangent plane matches three pieces of first-order information: the height $f(a, b)$, the x -direction slope $f_x(a, b)$, and the y -direction slope $f_y(a, b)$. A useful normal vector to the tangent plane is

$$\langle f_x(a, b), f_y(a, b), -1 \rangle.$$

3 Worked example: tangent plane to an elliptic paraboloid

Example 3.1: Tangent plane to $z = 3x^2 + y^2$

Find the tangent plane to

$$z = f(x, y) = 3x^2 + y^2$$

at the point $(1, 2, 7)$.

First check the height:

$$f(1, 2) = 3(1)^2 + (2)^2 = 7.$$

The partial derivatives are

$$f_x(x, y) = 6x, \quad f_y(x, y) = 2y.$$

At $(1, 2)$,

$$f_x(1, 2) = 6, \quad f_y(1, 2) = 4.$$

Therefore the tangent plane is

$$z - 7 = 6(x - 1) + 4(y - 2),$$

or

$$z = 6x + 4y - 7.$$

Thus the linearization is

$$L(x, y) = 6x + 4y - 7.$$

To estimate $f(1.1, 1.9)$,

$$L(1.1, 1.9) = 6(1.1) + 4(1.9) - 7 = 7.2.$$

The exact value is

$$f(1.1, 1.9) = 3(1.1)^2 + (1.9)^2 = 7.24.$$

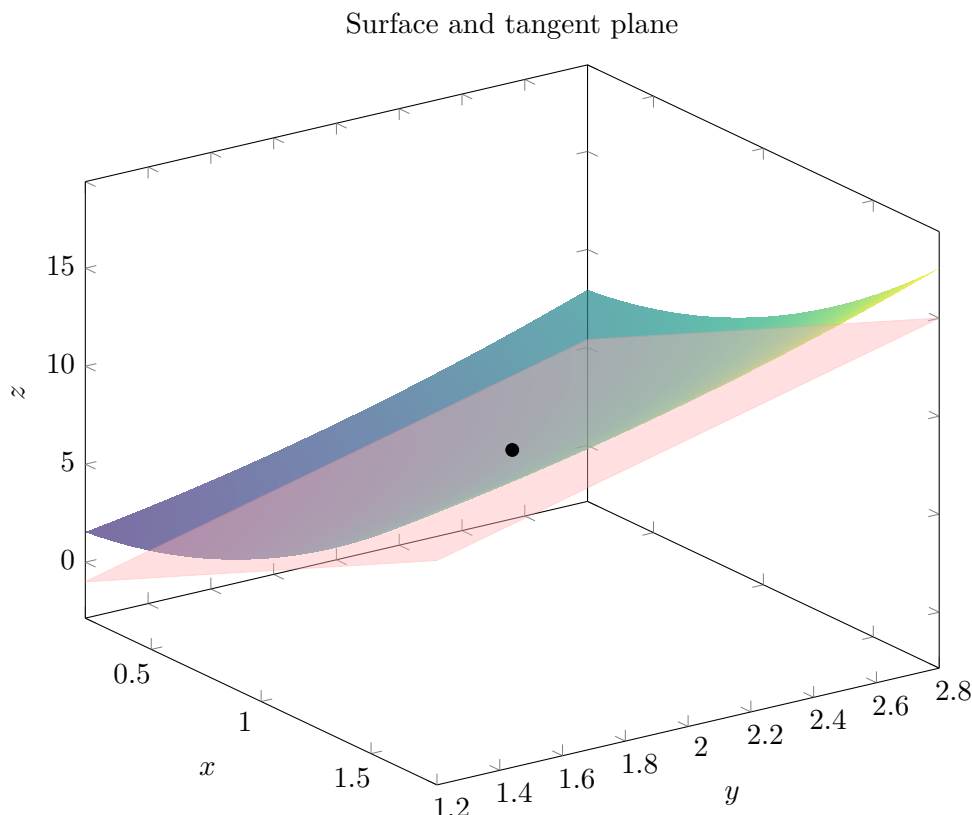


Figure 2: The graph of $z = 3x^2 + y^2$ and its tangent plane $z = 6x + 4y - 7$ at $(1, 2, 7)$.

4 Linear approximation in gradient form

Let

$$\Delta x = x - a, \quad \Delta y = y - b, \quad \Delta \mathbf{x} = \langle \Delta x, \Delta y \rangle.$$

Then

$$f(x, y) - f(a, b) \approx \nabla f(a, b) \cdot \Delta \mathbf{x}.$$

Equivalently,

$$\Delta z \approx \nabla f(a, b) \cdot \Delta \mathbf{x}.$$

For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the same formula is

$$f(\mathbf{x}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}).$$

Example 4.1: Gradient-dot-displacement estimate

Suppose

$$f(2, 1) = 13, \quad \nabla f(2, 1) = \langle 12, 2 \rangle.$$

Estimate $f(2.1, 0.9)$.

The input displacement is

$$\Delta \mathbf{x} = \langle 2.1 - 2, 0.9 - 1 \rangle = \langle 0.1, -0.1 \rangle.$$

Therefore

$$\Delta z \approx \langle 12, 2 \rangle \cdot \langle 0.1, -0.1 \rangle = 12(0.1) + 2(-0.1) = 1.$$

So

$$f(2.1, 0.9) \approx 13 + 1 = 14.$$

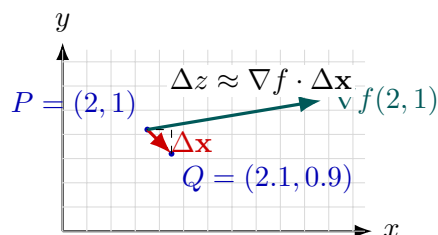


Figure 3: Linear approximation uses the dot product of the gradient and the displacement vector.

5 Approximation error and curvature

The linearization keeps only first-order information. The error comes from second-order and higher-order behavior. For

$$f(x, y) = 3x^2 + y^2,$$

the tangent plane at $(1, 2)$ is

$$L(x, y) = 6x + 4y - 7.$$

The error is

$$E(x, y) = f(x, y) - L(x, y) = 3(x - 1)^2 + (y - 2)^2.$$

This is always nonnegative because the surface is convex and lies above its tangent plane. The error is small near $(1, 2)$ and grows quadratically as we move away.

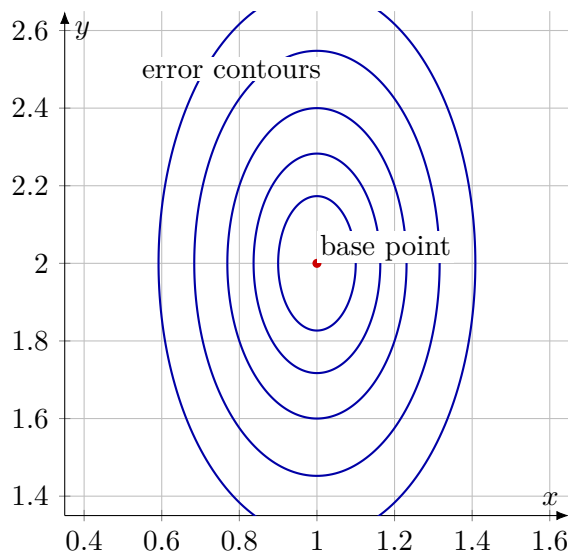


Figure 4: The linearization error is small near the base point and grows quadratically as we move away.

Connection to later approximation

Taylor approximation makes this precise. The second-order term involves the Hessian matrix. Linear approximation is first-order Taylor approximation; quadratic approximation is second-order Taylor approximation.

6 Differentiability

A function $f(x, y)$ is *differentiable* at (a, b) if its change can be written as a linear part plus an error that becomes negligible compared with the size of the input change. One way to state this is

$$\Delta z = f_x(a, b)\Delta x + f_y(a, b)\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y,$$

where

$$\varepsilon_1 \rightarrow 0, \quad \varepsilon_2 \rightarrow 0$$

as $(\Delta x, \Delta y) \rightarrow (0, 0)$. Equivalently,

$$f(a + \Delta x, b + \Delta y) = f(a, b) + f_x(a, b)\Delta x + f_y(a, b)\Delta y + \text{small error}.$$

The important idea is not just that the partial derivatives exist. The important idea is that together they produce a good linear model near the point.

Partial derivatives alone are not enough

A function may have partial derivatives at a point but fail to be differentiable there. Differentiability is stronger than the existence of coordinate-direction slopes.

Sufficient condition for differentiability

If f_x and f_y exist near (a, b) and are continuous at (a, b) , then f is differentiable at (a, b) .

7 A cautionary example

Consider

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2 + y^2}}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

At the origin, the coordinate partial derivatives exist:

$$f_x(0, 0) = 0, \quad f_y(0, 0) = 0.$$

So the candidate linearization is $L(x, y) = 0$. However, along the line $y = x$,

$$f(x, x) = \frac{x^2}{\sqrt{2x^2}} = \frac{|x|}{\sqrt{2}}.$$

This is not negligible compared with the distance $\sqrt{x^2 + x^2} = \sqrt{2}|x|$. Therefore the function is not differentiable at the origin.

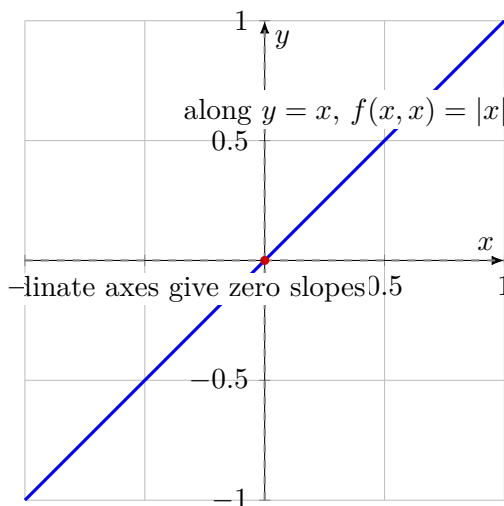


Figure 5: A function can have coordinate partial derivatives at a point but still fail to be differentiable there.

8 Tangent planes to level surfaces

Many important surfaces are not given as graphs $z = f(x, y)$. Instead, they are given implicitly as level surfaces

$$F(x, y, z) = c.$$

Suppose $P = (a, b, c_0)$ is a point on the level surface. If

$$\nabla F(P) \neq \mathbf{0},$$

then the gradient vector is normal to the level surface at P . Therefore the tangent plane is

$$\nabla F(P) \cdot \langle x - a, y - b, z - c_0 \rangle = 0.$$

In coordinates, if

$$\nabla F(P) = \langle F_x(P), F_y(P), F_z(P) \rangle,$$

then the tangent plane is

$$F_x(P)(x - a) + F_y(P)(y - b) + F_z(P)(z - c_0) = 0.$$

If a curve $\mathbf{r}(t)$ lies on the level surface, then $F(\mathbf{r}(t)) = c$. Differentiating with respect to t gives

$$\nabla F(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 0.$$

Thus the gradient is perpendicular to every tangent direction on the surface.

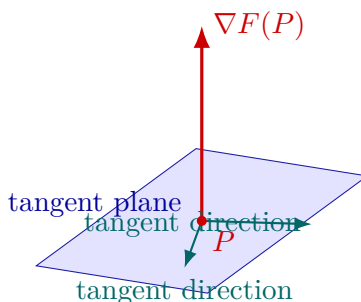


Figure 6: For a level surface, the gradient $\nabla F(P)$ is normal to the tangent plane.

9 Worked example: linearization and tangent plane to a level surface

Example 9.1: Linearization and tangent plane for a level surface

Let

$$F(x, y, z) = x^3 + y^2 e^z + xyz$$

and let $P = (1, 2, 0)$.

First evaluate the function:

$$F(1, 2, 0) = 1^3 + 2^2 e^0 + 1 \cdot 2 \cdot 0 = 5.$$

The partial derivatives are

$$F_x = 3x^2 + yz,$$

$$F_y = 2ye^z + xz,$$

$$F_z = y^2 e^z + xy.$$

At $P = (1, 2, 0)$,

$$F_x(P) = 3, \quad F_y(P) = 4, \quad F_z(P) = 6.$$

So the linearization is

$$L(x, y, z) = 5 + 3(x - 1) + 4(y - 2) + 6z.$$

Equivalently,

$$L(x, y, z) = 3x + 4y + 6z - 6.$$

Now estimate:

$$F(1.1, 1.9, 0) \approx L(1.1, 1.9, 0) = 3(1.1) + 4(1.9) - 6 = 4.9.$$

The level surface through P is $F(x, y, z) = 5$. Its tangent plane is

$$3(x - 1) + 4(y - 2) + 6z = 0,$$

or

$$3x + 4y + 6z - 11 = 0.$$

10 Total differentials

For a differentiable function $z = f(x, y)$, define dx and dy as independent small changes in the input variables. The *total differential* is

$$dz = f_x(x, y) dx + f_y(x, y) dy.$$

At a specific point (a, b) , this becomes

$$dz = f_x(a, b) dx + f_y(a, b) dy.$$

The differential is the linear approximation to the actual change:

$$\Delta z \approx dz.$$

For a function of three variables $w = f(x, y, z)$,

$$dw = f_x dx + f_y dy + f_z dz.$$

More generally, for $f : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$df = \nabla f(\mathbf{x}) \cdot d\mathbf{x}.$$

Why differentials matter

Deltas such as Δx and Δy represent actual changes. Differentials such as dx and dy are independent input changes used in the linear model. The differential dz is the predicted output change from the tangent plane.

11 Error propagation and sensitivity analysis

In applied problems, inputs are often measured with error. Linear approximation estimates how input error affects output error. Suppose

$$z = f(x, y)$$

and the possible measurement errors are

$$|dx| \leq \epsilon_x, \quad |dy| \leq \epsilon_y.$$

Then a first-order estimate for the output error is

$$|dz| \leq |f_x(x, y)|\epsilon_x + |f_y(x, y)|\epsilon_y.$$

If dx and dy are random errors with covariance matrix Σ , then the approximate variance of the output is

$$\text{Var}(df) \approx \nabla f(\mathbf{x})^T \Sigma \nabla f(\mathbf{x}).$$

This is the core idea behind the delta method in statistics.

Example 11.1: Volume of a cylinder

The volume of a cylinder is

$$V(r, h) = \pi r^2 h.$$

The differential is

$$dV = 2\pi r h dr + \pi r^2 dh.$$

At $r = 3$ and $h = 10$,

$$dV = 60\pi dr + 9\pi dh.$$

If $dr = 0.02$ and $dh = -0.05$, then

$$dV \approx 60\pi(0.02) + 9\pi(-0.05) = 0.75\pi \approx 2.36.$$

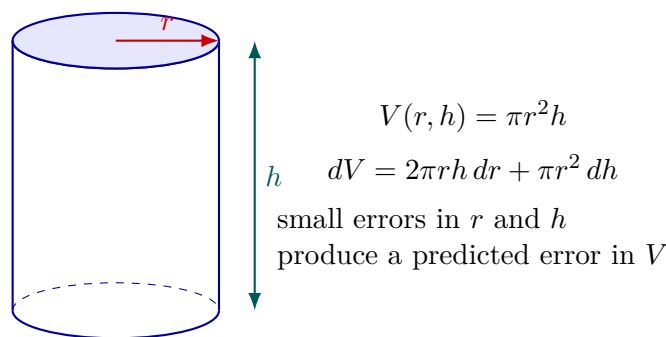


Figure 7: Linear error propagation predicts how measurement errors in radius and height affect cylinder volume.

12 Linearization as a local model in statistics and machine learning

Many statistical and machine-learning models depend on parameters. A simple example is a two-parameter linear prediction model

$$\hat{y} = \beta_0 + \beta_1 x.$$

A more complicated model may have many parameters:

$$\hat{y} = f(x; \theta_1, \theta_2, \dots, \theta_n).$$

Near a parameter vector θ_0 , linear approximation gives

$$f(x; \theta) \approx f(x; \theta_0) + \nabla_{\theta} f(x; \theta_0) \cdot (\theta - \theta_0).$$

This is the local linear model used in sensitivity analysis, uncertainty propagation, first-order optimization, local approximation of nonlinear regression models, and gradient-based learning.

Local linearization of a sigmoid

The logistic sigmoid is

$$\sigma(t) = \frac{1}{1 + e^{-t}}.$$

If $t = \beta_0 + \beta_1 x$, then

$$p(x; \beta_0, \beta_1) = \sigma(\beta_0 + \beta_1 x).$$

For fixed x , this is a function of the parameters β_0 and β_1 . Its gradient tells how the predicted probability changes when the parameters change.

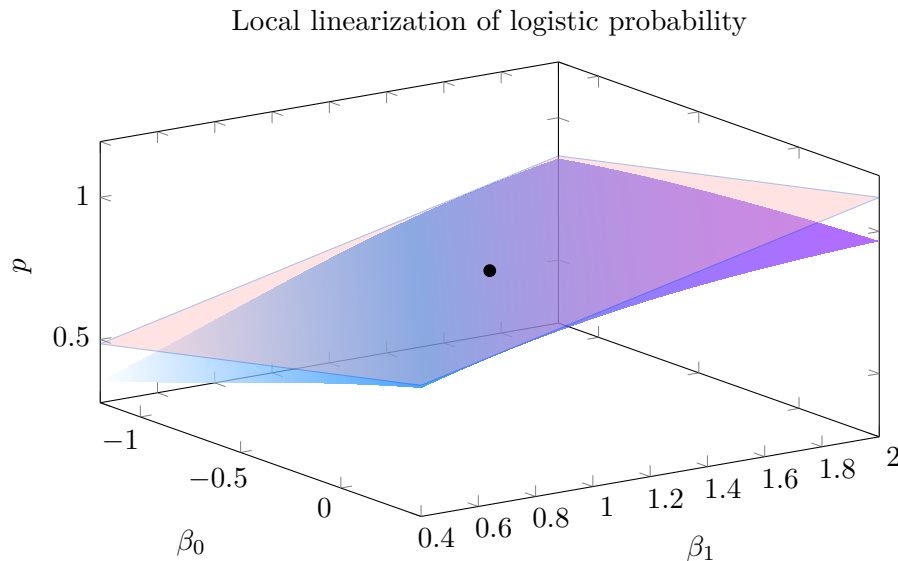


Figure 8: A nonlinear probability model is locally approximated by its tangent plane in parameter space.

13 Numerical linearization from finite differences

In real applications, we may not have a symbolic formula for the gradient. We may only have a black-box function that can be evaluated. Then we can approximate the gradient numerically. For a function $f(x, y)$, central finite differences give

$$f_x(a, b) \approx \frac{f(a + h, b) - f(a - h, b)}{2h},$$

$$f_y(a, b) \approx \frac{f(a, b + h) - f(a, b - h)}{2h}.$$

Then the numerical linearization is

$$L(x, y) = f(a, b) + \hat{f}_x(a, b)(x - a) + \hat{f}_y(a, b)(y - b).$$

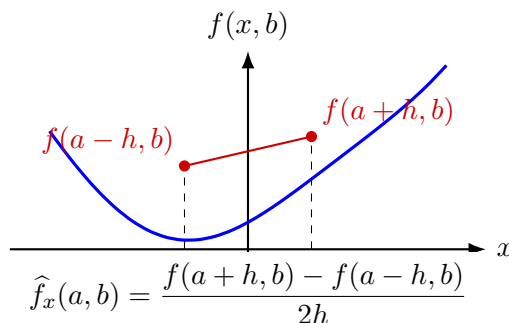


Figure 9: Finite differences approximate the gradient used in a local linear model.

14 Vector-valued and matrix viewpoints

The scalar-valued formula

$$f(\mathbf{x}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a})$$

has a natural extension to vector-valued functions. If

$$\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m,$$

then the best first-order approximation is

$$\mathbf{F}(\mathbf{x}) \approx \mathbf{F}(\mathbf{a}) + J_{\mathbf{F}}(\mathbf{a})(\mathbf{x} - \mathbf{a}),$$

where $J_{\mathbf{F}}(\mathbf{a})$ is the *Jacobian matrix*. The key idea is that derivatives in many variables are linear maps. The gradient-dot-product formula is the scalar-output version of this principle.

15 Common mistakes

Mistake 1: using the wrong base point

The tangent plane formula must use the point of tangency (a, b) and the height $f(a, b)$. Do not plug in the nearby point where you want the estimate.

Mistake 2: mixing up graph surfaces and level surfaces

For a graph $z = f(x, y)$, use

$$z - f(a, b) = f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

For a level surface $F(x, y, z) = c$, use

$$\nabla F(P) \cdot \langle x - a, y - b, z - c_0 \rangle = 0.$$

Mistake 3: thinking partial derivatives guarantee differentiability

Partial derivatives measure coordinate-direction slopes. Differentiability means there is a good linear approximation in all directions at once.

16 Summary of key formulas

For $z = f(x, y)$:

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Tangent plane to $z = f(x, y)$:

$$z - f(a, b) = f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Gradient form:

$$f(x, y) - f(a, b) \approx \nabla f(a, b) \cdot \langle x - a, y - b \rangle.$$

Total differential:

$$dz = f_x dx + f_y dy.$$

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$:

$$f(\mathbf{x}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}).$$

Tangent plane to $F(x, y, z) = c$ at $P = (a, b, c_0)$:

$$F_x(P)(x - a) + F_y(P)(y - b) + F_z(P)(z - c_0) = 0.$$

17 Practice problems

Conceptual problems

1. In your own words, explain why a tangent plane is a local linear model.
2. What information does the tangent plane to $z = f(x, y)$ match at the base point?
3. Explain why the gradient of F is normal to the level surface $F(x, y, z) = c$.
4. Explain the difference between Δz and dz .
5. Why can a function have partial derivatives but fail to be differentiable?

Skill problems

6. Find the tangent plane to $z = x^2 + 3y^2$ at $(1, -1, 4)$.
7. Find the linearization of $f(x, y) = e^{xy} + x^2y$ at $(1, 0)$.
8. Use a linear approximation to estimate $\sqrt{(4.1)^2 + (2.9)^2}$.
9. Suppose $f(3, 2) = 10$ and $\nabla f(3, 2) = \langle -2, 5 \rangle$. Estimate $f(2.9, 2.2)$.
10. Find dz for $z = x^2 \sin y + e^{xy}$.

Level-surface problems

11. Find the tangent plane to $x^2 + y^2 + z^2 = 14$ at $(1, 2, 3)$.
12. Find the tangent plane to $x^3 + y^2e^z + xyz = 5$ at $(1, 2, 0)$.
13. Find the tangent plane to $\ln(x + y + z) = 0$ at $(1, 0, 0)$.
14. Find a normal vector to the level surface $F(x, y, z) = x^2y + yz^2$ at $(1, 2, 1)$.
15. Find the linearization of $F(x, y, z) = x^2 + y^2 + z^2$ at $(1, 2, 2)$.

Applied and computational problems

16. The volume of a rectangular box is $V = xyz$. Use differentials to estimate the change in volume when $(x, y, z) = (10, 5, 2)$ changes by $(dx, dy, dz) = (0.1, -0.05, 0.02)$.
17. The BMI formula is $B(W, H) = 703W/H^2$, where W is weight in pounds and H is height in inches. Use differentials to estimate how small errors in W and H affect BMI.
18. For $f(x, y) = \sin(xy) + x^2$, use finite differences to estimate the tangent plane at $(1, 1)$.
19. For a prediction model $p(\beta_0, \beta_1) = 1/(1 + e^{-(\beta_0 + 2\beta_1)})$, find the linearization at $(0, 1)$.
20. Choose a nonlinear function of two variables, plot the surface and tangent plane, and describe where the approximation is accurate.

18 Python lab and interactive page

The Python lab for this chapter should ask students to build local linear models from symbolic formulas, numerical finite differences, and data-based approximations. Recommended lab activities:

1. visualize the surface $z = 3x^2 + y^2$ and its tangent plane at $(1, 2)$;
2. compare exact values with linearized values on a grid;
3. compute and visualize approximation error;
4. estimate gradients numerically using finite differences;
5. find tangent planes to level surfaces using gradients;
6. simulate error propagation for measurement uncertainty;
7. explore local linearization of a logistic probability model.

Resources:

- Notebook lab: Chapter 9 linear approximation lab.
- Interactive HTML page: `chapter-09.html`.

19 Chapter summary

Linear approximation is the first major synthesis point of multivariable calculus. Partial derivatives become a gradient. The gradient becomes a local linear model. The local linear model becomes a tangent plane, a differential, a sensitivity estimate, and a computational approximation.

For graphs $z = f(x, y)$, the tangent plane is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

For level surfaces $F(x, y, z) = c$, the tangent plane is determined by the normal vector $\nabla F(P)$:

$$\nabla F(P) \cdot (\mathbf{x} - P) = 0.$$

In applied mathematics and statistics, this chapter is the foundation for error propagation, the delta method, local optimization, and first-order approximation of nonlinear models. In machine learning and AI, it is the idea behind using gradients to approximate and improve complicated models one small step at a time.